Erald Justin Sanchez

0930-556-4794

eraldjustinsanchez@gmail.com

San Carlos City, Pangasinan

https://github.com/eraldzz

https://portfolio-wine-sigma-28.vercel.app/

CAREER OBJECTIVE

Results-oriented Computer Engineering graduate with specialized certifications in Data Engineering (IBM) and Cloud Infrastructure (Oracle). Proficient in building automated ETL pipelines and Web Applications using Python, JavaScript, and SQL. Strong foundational knowledge in AI/Computer Vision, IoT integration, and cloud security.

TECHNICAL SKILLS

- Languages: Python (Pandas, BeautifulSoup), JavaScript (ES6+), Java, C++, SQL.
- Data & Cloud: ETL Pipelines, Oracle Cloud Infrastructure (OCI), PostgreSQL, MySQL.
- Tools & Security: Git, Raspberry Pi, Arduino, OpenCV, REST APIs, Vercel, IAM, VCN.

RELEVANT EXPERIENCE & PROJECTS

World Banks Data Engineering Pipeline | IBM Certification Project

- Developed a Python-based ETL pipeline to automate the extraction of financial data from web sources via BeautifulSoup.
- Transformed raw datasets using Pandas for data cleaning and loaded structured data into SQLite for SQL-based querying.
- Implemented logging and error-handling scripts to ensure data integrity throughout the pipeline.

Personal Web Portfolio | Web Development

- Developed a custom web portfolio using Vanilla HTML 5, CSS3, and JavaScript to showcase engineering and technical skills.
- Deployed and maintained web applications using Vercel, ensuring 100% uptime and responsive design of mobile and desktop.

AI- Driven Fire Detection & Suppression System | *IoT & Computer Vision*

- Developed a real-time detection system using Python and OpenCV on a Raspberry Pi to identify fire and smoke patterns via live camera feed.
- Integrated a dual-processor architecture, utilizing the Raspberry Pi for high-level image processing and an Arduino for low-level sensor management and actuator control.
- Engineered a suppression mechanism that triggers automated water pumps and emergency alerts upon high-confidence detection.

Automated Irrigation System | Embedded Systems Project

- Designed and programmed an Arduino-based control system to automate water distribution based on real-time soil moisture levels.
- Integrated soil moisture sensors and water pumps using C++, optimizing power consumption and water efficiency.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Foundations Associate — Oracle University
- Introduction to Data Engineering — IBM (Coursera), 2026
- Python for Data Science, AI and Development — IBM (Coursera), 2026
- Python Project for Data Engineering — IBM (Coursera), 2026

EDUCATION

Bachelor of Science in Computer Engineering
PHINMA - University of Pangasinan
Dagupan City, Pangasinan